

microservices hosted on Amazon Web Services (AWS). It uses a combination of Google Assistant, Amazon Lex and core IoT services to talk to a user and answer queries.

It has a camera in the front to recognise faces. It uses computer vision to identify a user and map it to the database, to fetch the recipe he or she likes the most.

This means if you walk in front of Arya, it will greet you by your name, read your favourite flavour and ask you if it should blend a cup of, say, *masala chai* (your typical order). Thus, Arya understands who it is speaking to and what a user typically prefers.

Arya is built using custom-designed electronics with patented technology to make a perfect cup of tea. It can be remotely diagnosed, as it sends data about ingredient quantities, system and network back to the central server.

Explaining the role of the IoT here, Akash Takyar, chief executive officer and co-founder, Arya, says, "Arya is an IoT device that is connected to Google Cloud IoT. It sends usage data and collects feedback. Based on user feedback, Arya adjusts the ingredients to improve the recipe-building process. As it is connected, the user can also place an order remotely and Arya will make the perfect cup of *chai*."

Arya's core technology is a combination of computer vision, voice, cloud APIs and mechanics that makes a cup of tea using user instructions.

Talking about Arya's journey and its road map ahead, Takyar adds, "Getting a good cup of tea in offices is tough. Most employees end up either at street vendors or expensive cafés. Also, tea made with tea bags does not give the flavour of brewed, home-like tea.

"In 2016, a team of AI experts, industrial automation and IoT engineers collaborated to solve this problem with the IoT. Two years and six iterations later, a production-ready robot was created that could be installed in offices, to make custom *chai* using real ingredients.

"Arya will soon be available at malls, cafés, airports and offices."

The IoT makes *chai* consumption error-free and transparent

While the entire world has resorted to technology for making everything error-free, why should the consumption of *chai* go amiss? There are tea joints like Chai Point that focus on making your cup of *chai* faultless and data savvy, thanks to the IoT.

The IoT- and Android-powered boxC.in machines use the IoT to throw error codes in case a discrepancy occurs in the machine. These codes lead to a ticketing mechanism, providing first-hand information about the error(s) to the repair-and-management team, who can come up with a quick solution. Not just saving time for customers, this IoT-enabled machine can also reflect data about every cup of tea consumed.

Speaking of the benefits of boxC.in, Shirish Surti, vice president - engineering, Chai Point, says, "Several organisations have installed boxC.in machines to serve *chai* with fresh milk to their employees, to avoid using pre-mix-based beverages. These machines provide employees access to error-free digital invoicing and transparency in consumption, keeping a constant eye on the procedure."

Shedding light on how boxC.in works, Surti adds, "It was in 2016 that we flagged off boxC.in—one year later came the Android version. We have a command centre that watches the machine's condition, consumption patterns, stock status and more. Its IoT-powered feature enables it to generate reports and alert the command centre regarding any issue(s). We have more than 2000 boxC.in installations that are operating at present in the country."

Chai Monk brews *Meri Wali Chai* for all

Missing your mom's handmade malai *chai*? Chai Monk, an AI-based robot that makes customised tea, gets the same flavour right to your table consistently, irrespective of the Chaayos café you visit. It makes sure you get *Meri Wali Chai* across all locations. The robot's uniqueness

lies in the handmade way it brews a cup of tea for you.

Explaining the part that the IoT plays, Nitin Saluja, founder and chief executive officer, Chaayos, says, "Chai Monk is an IoT-based robot. When you place an order, the Monk automatically receives the details and starts brewing the *chai*. It can customise ingredients like milk, sugar, tea leaves and add-ons, by simply placing the order at the POS machine. It has completely automated the process of *chai*-making, bringing consistency across all Chaayos cafés. As we expanded our presence to more cities and locations, we wanted to ensure that the experience remains seamless and consistent with every order."

At Chaayos, technology is the backbone. "It powers our real-time, cloud-based POS system, supply chain module, delivery business and Web app, analytics, training and, of course, Chai Monk. Technology brings efficiency and speed to the system and helps reduce man and material wastage. It also helps in identifying what works best for our customers. All our menu additions are based on customer data, and we have been able to tailor our customer experience accordingly," says Saluja.

He adds, "Our café teams are trained through an online video-based training module. The entire process of training and certification is completely automated and can be monitored to ensure maximum effectiveness. Kettle, Chaayos' consumer insight tool, enables live inventory status at our café locations. It helps keep order cancellation and wastage to a bare minimum."

Needless to say, the IoT is no more limited to gadgets and appliances. It has already sneaked to our cups and plates. It is great news for tech addicts to be able to smell technology in their daily cup of *chai*.

Apart from Arya, Chai Point and Chaayos, there are several other India-based tea joints such as Tea Halt and Madras Cafe that are making use of advanced technology like the IoT, AI and robotics to blend the perfect cup of *chai* for you. Next time you visit a tea outlet, make sure to not miss these technovations. **EFY**

—Sreemoyee Chatterjee, EFY